

01-04 · ALGEBRA

# Linear equations & lines

Translate → solve → return the requested quantity.

## 01 · Equations: isolate what the question asks

Distribute, combine like terms, then undo operations on both sides. Translate words before calculating: “5 less than twice  $x$ ” is  $2x - 5$ .

**Example** If  $3(2x - 5) = 21$ , then  $2x - 5 = 7$ ,  $2x = 12$ , and  $x = 6$ . If asked for  $2x$ , answer **12**.

**After simplifying:** a true statement such as  $0 = 0$  means all allowed values work; a false statement such as  $0 = 5$  means no solution.

## 02 · Lines: slope is change per unit

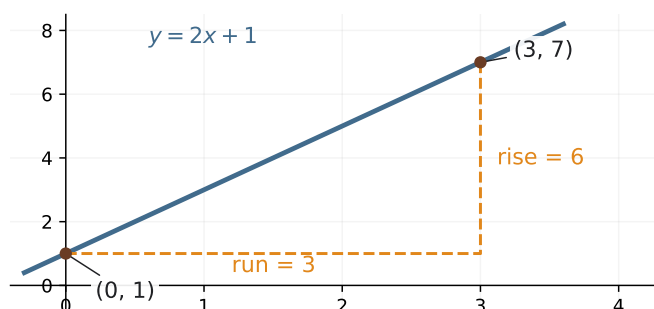
$$y = mx + b$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

**m:** slope. **b:** y-intercept.

**Point-slope:**  $y - y_1 = m(x - x_1)$ .

In the graph, slope =  $6 \div 3 = 2$ . The y-intercept is **1**.



**Parallel:** distinct nonvertical lines are parallel when their slopes match. Distinct vertical lines are also parallel.

**Perpendicular:** slopes multiply to  $-1$ , when both are defined. A horizontal line and a vertical line are perpendicular.

**Intercepts:** set  $y = 0$  for an x-intercept; set  $x = 0$  for a y-intercept. Vertical  $x = c$  has undefined slope; horizontal  $y = c$  has slope 0.

## 03 · Systems: a solution must satisfy both

| Relationship                     | What happens                              |
|----------------------------------|-------------------------------------------|
| Different slopes                 | One intersection → one solution           |
| Same slope, different intercepts | Parallel lines → no solution              |
| Equivalent equations             | The same line → infinitely many solutions |

**Example**  $x + y = 7$  and  $x - y = 1$ . Add:  $2x = 8$ , so  **$(x, y) = (4, 3)$** .

Use substitution or elimination. Desmos: graph both and read the intersection (p. 9).

## 04 · Inequalities: one sign rule matters

**Reverse the inequality when multiplying or dividing by a negative.** All other standard solving steps keep its direction.

**Example**  $-2x > 6 \rightarrow x < -3$ . For  $1 < 2x + 3 \leq 7$ , solve all three parts:  **$-1 < x \leq 2$** .

**AND:** overlap of conditions. **OR:** either condition works. For parameter questions, consider a zero coefficient before dividing by it.

05-08 · ADVANCED MATH

# Quadratics & their graphs

Match the question to a useful form before doing extra algebra.

## 01 · Choose the form that reveals the answer

| Form · $a \neq 0$         | Read directly                                           |
|---------------------------|---------------------------------------------------------|
| $y = ax^2 + bx + c$       | y-intercept: <b>c</b>                                   |
| $y = a(x - h)^2 + k$      | Vertex: <b>(h, k)</b> ; axis: <b><math>x = h</math></b> |
| $y = a(x - r_1)(x - r_2)$ | Zeros: <b><math>r_1</math> and <math>r_2</math></b>     |

**$a > 0$ :** opens up; the vertex is a minimum.  **$a < 0$ :** opens down; the vertex is a maximum. Larger  $|a|$  means a narrower parabola.

## 02 · Read the whole graph

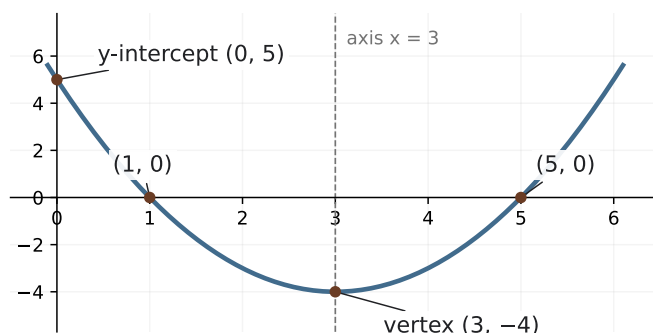
$$\begin{aligned} y &= (x - 3)^2 - 4 \\ &= x^2 - 6x + 5 \\ &= (x - 1)(x - 5) \end{aligned}$$

**Vertex:** (3, -4)

**Zeros:** 1 and 5

**y-intercept:** 5

**Minimum value:** -4

The minimum is the **y-value**; it occurs when  $x = 3$ .


## 03 · Solve or count real solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad D = b^2 - 4ac$$

| Discriminant | Distinct real solutions |
|--------------|-------------------------|
| $D > 0$      | Two                     |
| $D = 0$      | One repeated root       |
| $D < 0$      | None                    |

**Example**  $x^2 - 6x + k = 0$  has one real solution when  $36 - 4k = 0$ , so  **$k = 9$** .

For a line–parabola system, substitute first, then apply the discriminant to the resulting quadratic.

## 04 · Factor, complete the square, or use roots

**Zero-product rule:** if  $AB = 0$ , then  $A = 0$  or  $B = 0$ . First move everything to one side.

$$\begin{aligned} u^2 - v^2 &= (u - v)(u + v) \\ (u \pm v)^2 &= u^2 \pm 2uv + v^2 \end{aligned}$$

**Complete the square:**

$$x^2 + 6x + 5 = (x + 3)^2 - 4.$$

For roots of  $ax^2 + bx + c = 0$ :

**sum** =  $-b/a$ ; **product** =  $c/a$ .

**Vertex x-coordinate:**  $-b/(2a)$ .

A graph suggests an answer. Algebra confirms an exact value or a “one solution” condition.

09–12 · ADVANCED MATH

# Powers, expressions & functions

Use structure. Keep restrictions. Check any squared equation.

## 01 · Powers & radicals

| Rule                                            | Example               |
|-------------------------------------------------|-----------------------|
| $a^m \cdot a^n = a^{m+n}$                       | $x^3 \cdot x^4 = x^7$ |
| $a^m \div a^n = a^{m-n}; a \neq 0$              | $x^7 \div x^2 = x^5$  |
| $(a^m)^n = a^{mn}$                              | $(x^3)^2 = x^6$       |
| $a^{-n} = 1/a^n; a \neq 0$                      | $2^{-3} = 1/8$        |
| $a^0 = 1; a \neq 0$                             | $7^0 = 1$             |
| $a^{m/n} = (\sqrt[n]{a})^m, \text{ for } a > 0$ | $27^{2/3} = 9$        |

**Real-number checks:** an even root needs a nonnegative radicand.  $\sqrt{(x^2)} = |x|$ . Also,  $\sqrt{(a + b)}$  does not generally equal  $\sqrt{a} + \sqrt{b}$ .

## 03 · Functions: input versus output

**f(3):** substitute 3 for every x. **f(x) = 3:** find the input(s) giving output 3. **f(g(x)):** evaluate the inside function first.

**Example** If  $f(x) = x^2 + 2x + 3$ , then  $f(3) = 18$ . If  $g(x) = x + 1$ , then  $f(g(2)) = f(3) = 18$ .

**Domain:** allowed inputs. **Range:** resulting outputs. In a piecewise function, use the rule whose condition includes the given input.

## 04 · Transformations: inside changes feel reversed

$y = f(x - h) + k$  shifts a graph **right h**, **up k** when h and k are positive.

$-f(x)$ : reflect across the x-axis.

$f(-x)$ : reflect across the y-axis.

$c f(x)$ : multiply all y-values by c.

$f(cx)$ : horizontal scale  $1/|c|$ ,  $c \neq 0$ ; if  $c < 0$ , also reflect.

## 02 · Fractions, absolute values & restrictions

**Cancel factors, never terms:**

$(x^2 - 9)/(x - 3) = x + 3$ , but  $x \neq 3$  still applies.

**To add fractions:** use a common denominator. **To solve:** clear denominators, then check the original equation.

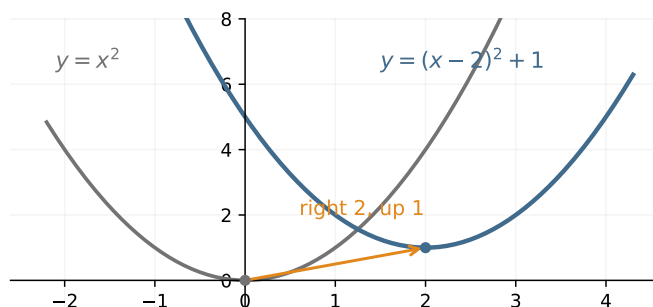
**Example**  $\sqrt{(x + 2)} = x$  gives candidates 2 and -1 after squaring. Only 2 works in the original.

**|x - a|** is the distance from x to a.

$|x - 2| = 5 \rightarrow x = -3 \text{ or } 7$ .

$|x - 2| < 5 \rightarrow -3 < x < 7$ .

A denominator cannot be zero. Keep original restrictions even after simplifying.



**Extra polynomial tool:** dividing  $P(x)$  by  $x - r$  leaves remainder  $P(r)$ . If  $P(r) = 0$ , then  $x - r$  is a factor.

13–16 · PROBLEM-SOLVING

## Percentages, ratios & models

Write the relationship first; attach units to your answer.

### 01 · Percentages: use a multiplier

| Situation                | Setup                                                    |
|--------------------------|----------------------------------------------------------|
| p% of a quantity         | Multiply by $p/100$                                      |
| Increase by p%           | Multiply by $1 + p/100$                                  |
| Decrease by p%           | Multiply by $1 - p/100$                                  |
| Percent change           | $(\text{new} - \text{old}) \div \text{old} \times 100\%$ |
| Find the original amount | Divide the final amount by its multiplier                |

**Example** After a 20% discount, a price is \$72. Original price:  $72 \div 0.80 = \$90$ .

**Successive changes multiply:** +20%, then -20% gives  $1.2 \times 0.8 = 0.96$ , a 4% decrease. From 40% to 50% is +10 percentage points, a 25% relative increase.

### 02 · Ratios, units & rates

If  $A : B = 2 : 3$ , write  **$A = 2k$ ,  $B = 3k$** . A is  $2/5$  of the total, not  $2/3$ .

**Example** A total of 40 split in the ratio 2 : 3 gives **16 and 24**.

distance = rate  $\times$  time

average speed = total distance  $\div$  total time

Convert with factors that cancel units. Square the conversion for area; cube it for volume.

**Example**  $1 \text{ ft}^2 = (12 \text{ in})^2 = \mathbf{144 \text{ in}^2}$ . Traveling 60 miles at 30 mph and 60 miles at 60 mph gives  $120 \div 3 = \mathbf{40 \text{ mph}}$ .

An inverse relationship has  $xy = k$ , or  $y = k/x$ . Do not average speeds unless the time intervals are equal.

### 04 · Linear versus exponential models

| Model                 | Meaning                                              |
|-----------------------|------------------------------------------------------|
| $y = b + mx$          | Adds $m$ per unit of $x$ ; $b$ is the starting value |
| $y = A(1 + r)^t$      | Grows by rate $r$ each period                        |
| $y = A(1 - r)^t$      | Decays by rate $r$ each period                       |
| $y = A \cdot b^{t/k}$ | Multiplies by $b$ every $k$ time units               |

**Example** A population starts at 800 and doubles every 3 years:  $P(t) = 800 \cdot 2^{t/3}$ . After 6 years: **3,200**.

For equally spaced inputs: constant differences suggest linear; constant ratios suggest exponential. Match the exponent's time unit to the question.

17–20 · PROBLEM-SOLVING

## Data, probability & studies

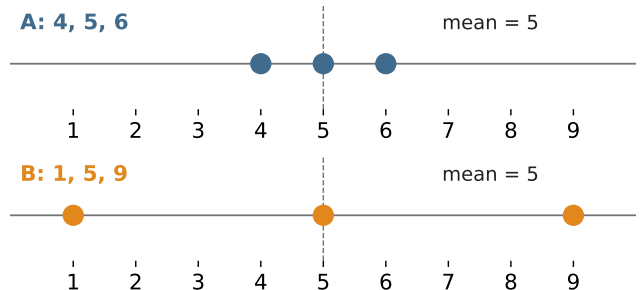
Read the group and units. Then choose the denominator.

### 01 · Center & spread

**Mean:**  $\text{sum} \div \text{count}$ . **Median:** middle value after sorting; average the middle two if the count is even. **Range:** maximum – minimum.

**Combined mean:** add the groups' totals, then divide by the combined count.

**Example** 10 students average 70; 20 average 85.  
Combined mean:  $(10 \times 70 + 20 \times 85)/30 = 80$ .



**Standard deviation measures spread.** B has the larger SD. Adding a constant to every value leaves SD unchanged; multiplying every value by  $c$  multiplies SD by  $|c|$ . Outliers usually affect the mean more than the median.

### 02 · Probability: choose the right denominator

$P(\text{event}) = \text{favorable outcomes} \div \text{total outcomes}$

For equally likely outcomes

|          | Club | Not club | Total |
|----------|------|----------|-------|
| Grade 11 | 12   | 18       | 30    |
| Grade 12 | 8    | 12       | 20    |
| Total    | 20   | 30       | 50    |

**Example** Given that a student is in grade 11,  $P(\text{club}) = 12/30 = 2/5$ . Restrict the denominator to the given group.

**Complement:**  $P(\text{not } A) = 1 - P(A)$ .

**Independent events:**  $P(A \text{ and } B) = P(A)P(B)$ .

**Union:**  $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$ .

Without replacement, the available outcomes change after the first selection.

### 04 · Samples, experiments & margin of error

| Study feature                   | What it supports                                            |
|---------------------------------|-------------------------------------------------------------|
| Random sample from a population | Generalizing to that population, if sampling is appropriate |
| Random assignment to treatments | Cause-and-effect conclusions in a well-designed experiment  |
| Observational study only        | Association; other explanations may remain                  |

**Estimate  $\pm$  margin of error:**  $54\% \pm 3$  percentage points gives **51% to 57%**. With comparable design and confidence level, a larger random sample generally reduces margin of error.

A large sample cannot fix a biased sampling method.

### 03 · Read data before calculating

Check the title, axes, units, scale, and group. In a histogram, bars show intervals; a median may only be located within an interval.

**Scatterplot:** describe direction, form, strength, and outliers. A line of best fit predicts a typical value.

$\text{residual} = \text{actual} - \text{predicted}$

**Example** Actual value 14; prediction 11  $\rightarrow$  residual **+3**, a point above the model.

**Correlation does not establish causation.** Predictions far beyond the observed data may be unreliable.

21–24 · GEOMETRY

## Geometry & measurement

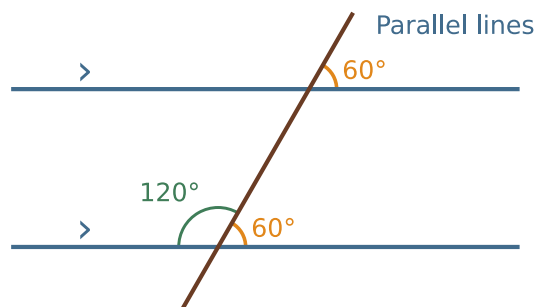
Use angle relationships and matching sides before calculating.

### 01 · Angle relationships

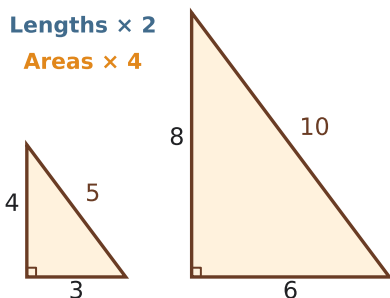
**Straight line:**  $180^\circ$ . **Around a point:**  $360^\circ$ . **Vertical angles:** equal.

With **parallel lines**, corresponding and alternate interior angles are equal; same-side interior angles sum to  $180^\circ$ .

**Triangle:** interior angles sum to  $180^\circ$ . An exterior angle equals the sum of the two remote interior angles.



### 02 · Similarity, congruence & scale



**Congruence:** same shape and size. Use SSS, SAS (included angle), ASA, AAS, or hypotenuse–leg for right triangles. S = side; A = angle. **AAA gives similarity only.**

**Similar triangles:** equal corresponding angles and proportional corresponding sides. Two matching angles establish similarity.

lengths  $\times k$   
areas  $\times k^2$   
volumes  $\times k^3$

**Example** Doubling every length makes area **4 times** as large and volume **8 times** as large.

The volume rule applies to similar solids. Match corresponding sides before making a proportion.

### 03 · Lengths & areas

| Shape / rule                        | Formula                                         |
|-------------------------------------|-------------------------------------------------|
| Triangle sides                      | Each side is less than the sum of the other two |
| Right triangle <span>R</span>       | $a^2 + b^2 = c^2$ ; c is the hypotenuse         |
| Rectangle / triangle <span>R</span> | $A = lw$ ; $A = \frac{1}{2}bh$                  |
| Parallelogram / trapezoid           | $A = bh$ ; $A = \frac{1}{2}(b_1 + b_2)h$        |
| Circle <span>R</span>               | $A = \pi r^2$ ; $C = 2\pi r$                    |
| Coordinate distance                 | $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$      |
| Midpoint                            | $((x_1 + x_2)/2, (y_1 + y_2)/2)$                |

A height is perpendicular to its base. Never infer equal lengths or angles just from a drawing.

### 04 · Volumes

| Solid <span>R</span> | Volume                 |
|----------------------|------------------------|
| Rectangular prism    | $lwh$                  |
| Cylinder             | $\pi r^2 h$            |
| Sphere               | $\frac{4}{3}\pi r^3$   |
| Cone                 | $\frac{1}{3}\pi r^2 h$ |
| Rectangular pyramid  | $\frac{1}{3}lwh$       |

**Any prism:**  $V = Bh$ .

**Any pyramid:**  $V = \frac{1}{3}Bh$ .

B is the area of the base.

R Formula supplied on the SAT reference sheet. Know how to identify each measurement and use squared or cubed units.

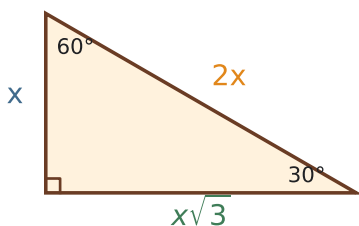
25–27 · GEOMETRY & TRIGONOMETRY

## Right triangles & trigonometry

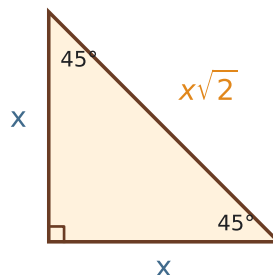
Identify the right angle, choose the reference angle, then label the sides.

### 01 · Special right triangles R

**30°–60°–90°**



**45°–45°–90°**

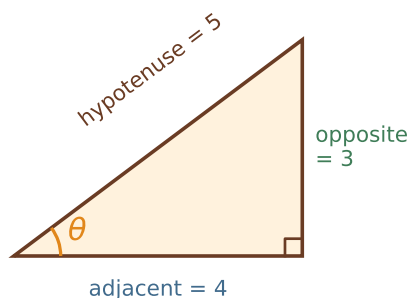


**30°–60°–90°:** find the short leg first.  
If the hypotenuse is 10,  $x = 5$ ; the long leg is  $5\sqrt{3}$ .

**45°–45°–90°:** the legs match.  
If the hypotenuse is 10, each leg is  $10/\sqrt{2} = 5\sqrt{2}$ .

**Match sides to opposite angles.** The hypotenuse is always opposite 90°. Rotating a triangle does not change the ratios.

### 02 · SOH–CAH–TOA



$$\begin{aligned}\sin \theta &= \text{opposite} / \text{hypotenuse} \\ \cos \theta &= \text{adjacent} / \text{hypotenuse} \\ \tan \theta &= \text{opposite} / \text{adjacent}\end{aligned}$$

For the pictured angle:  
 $\sin \theta = 3/5$ ;  $\cos \theta = 4/5$ ;  
 $\tan \theta = 3/4$ .

**Complementary acute angles:**  
 $\sin \theta = \cos(90^\circ - \theta)$ .

“Opposite” and “adjacent” change when you use the other acute angle.

### 03 · Exact values & radians

| Angle | Radians | $\sin \theta$ | $\cos \theta$ | $\tan \theta$ |
|-------|---------|---------------|---------------|---------------|
| 0°    | 0       | 0             | 1             | 0             |
| 30°   | $\pi/6$ | $1/2$         | $\sqrt{3}/2$  | $\sqrt{3}/3$  |
| 45°   | $\pi/4$ | $\sqrt{2}/2$  | $\sqrt{2}/2$  | 1             |
| 60°   | $\pi/3$ | $\sqrt{3}/2$  | $1/2$         | $\sqrt{3}$    |
| 90°   | $\pi/2$ | 1             | 0             | Undefined     |

**Convert:** degrees  $\times \pi/180$  = radians; radians  $\times 180/\pi$  = degrees. Thus  $180^\circ = \pi$  radians and  $360^\circ = 2\pi$  radians.

**Unit circle:** a point at angle  $\theta$  has coordinates  $(\cos \theta, \sin \theta)$ . Use the quadrant for signs. Always match the calculator’s angle mode to the question (p. 12).

28–31 · GEOMETRY + STRATEGY

## Circles & answer checks

Keep the center, radius, angle measure, and requested quantity distinct.

### 01 · Circle equations

$$(x - h)^2 + (y - k)^2 = r^2$$

**Center:**  $(h, k)$ . **Radius:**  $r$ , the square root of the right side. The signs inside the parentheses are opposite the center's coordinates.

**Example**  $x^2 + y^2 - 4x + 6y - 12 = 0$

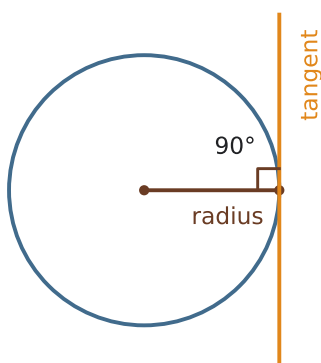
Complete both squares:  $(x - 2)^2 + (y + 3)^2 = 25$ .

Center **(2, -3)**; radius **5**.

Add the same amounts to both sides when completing squares. For the Desmos circle example, see p. 12.

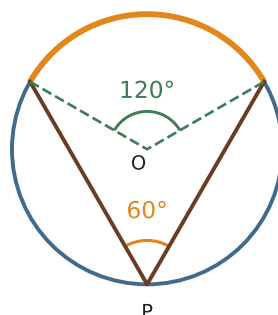
### 02 · Tangents & inscribed angles

#### Radius meets tangent at 90°



A radius is **perpendicular to the tangent** at the point of tangency. A tangent touches the circle at exactly one point.

#### Same arc: central = 2 × inscribed



An **inscribed angle is half its intercepted arc's measure**. An angle inscribed in a semicircle is 90°.

### 03 · Arcs & sectors

| Angle $\theta$ | Arc length $s$              | Sector area $A$              |
|----------------|-----------------------------|------------------------------|
| Degrees        | $(\theta/360) \cdot 2\pi r$ | $(\theta/360) \cdot \pi r^2$ |
| Radians        | $r\theta$                   | $\frac{1}{2}r^2\theta$       |

**Example** With  $r = 6$  and  $\theta = 60^\circ = \pi/3$ , the arc length is  **$2\pi$**  and the sector area is  **$6\pi$** .

Arc length is a distance; sector area is an area. Radian formulas require a radian angle.

### 04 · Final answer checks

- **Target:**  $x$ ,  $y$ , an expression, a length, or a count?
- **Restrictions:** zero denominators, radical conditions, valid time or length?
- **Exactness:** keep fractions and radicals until rounding is requested.
- **Completeness:** find all valid solutions, then answer exactly what is asked.
- **Entry:** for a student-produced response that accepts several values, enter only one. Do not enter a mixed number.

**Hard questions:** for “exactly one solution,” identify the structure first. A genuine linear equation needs a nonzero  $x$ -coefficient; a quadratic needs  $D = 0$ . Two identical lines have infinitely many solutions.

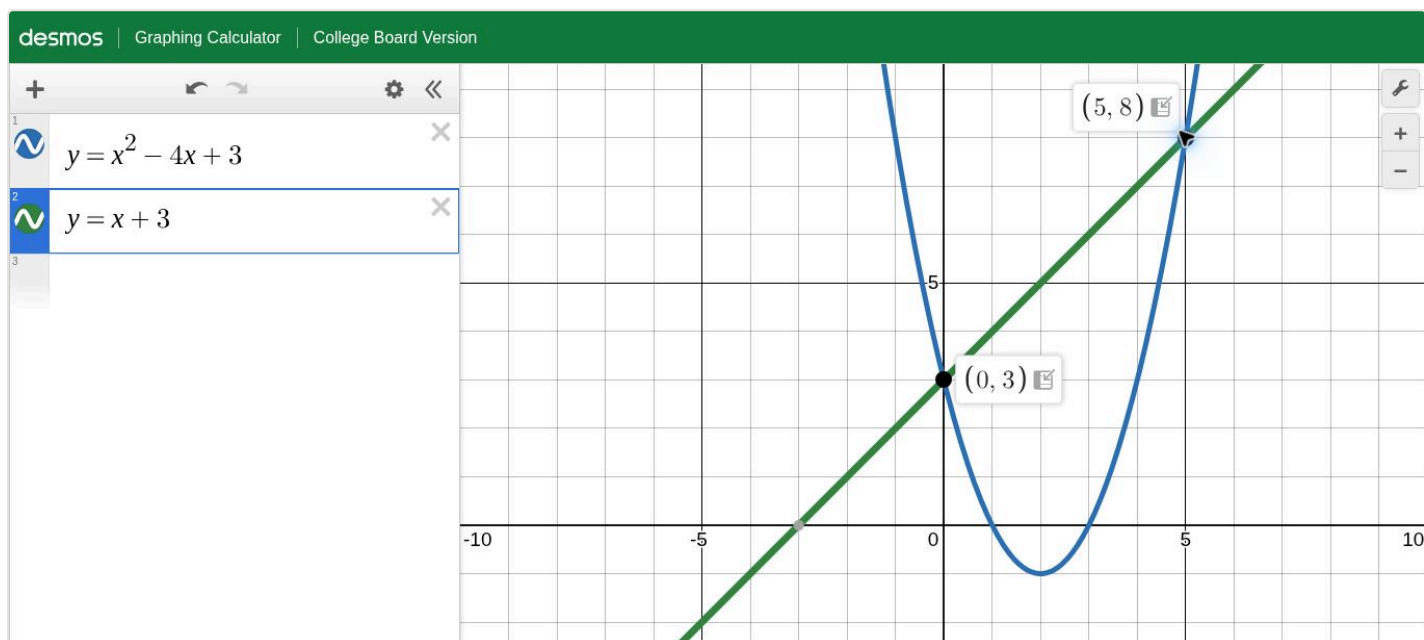
## CALCULATOR LAB · 1 OF 4

## Desmos · solve &amp; read graphs

Use the College Board calculator. Practice clicking the exact point you need.

## 01 · Solve an equation or system

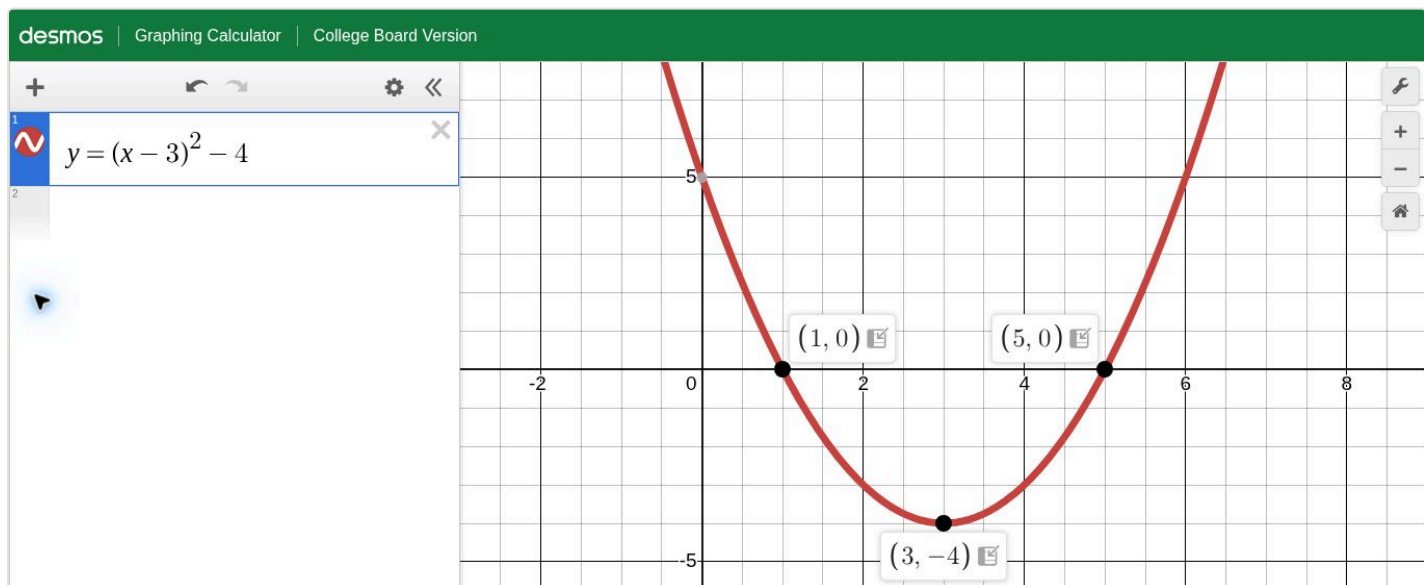
1. Enter each side as a separate graph. Here:  $y = x^2 - 4x + 3$  and  $y = x + 3$ .
2. Select a graph, then select each intersection to display its coordinates. Zoom or pan if needed.
3. Read the requested coordinate: the equation's x-solutions are **0 and 5**; the system solutions are **(0, 3) and (5, 8)**.



**Quick alternative:** for  $g(x) = 0$ , graph  $y = g(x)$  and select its x-intercepts. A y-intercept is not automatically a solution.

## 02 · Find zeros, a vertex, or an extreme value

1. Enter  $y = (x - 3)^2 - 4$ . Select the curve to reveal points of interest.
2. Select the intercepts and vertex. Here the zeros are **1 and 5**; the vertex is **(3, -4)**.
3. If asked for the minimum value, answer **-4**. If asked where it occurs, answer  **$x = 3$** .



**Try:** graph  $y = (x + 2)^2 + 1$ . Vertex  **$(-2, 1)$** ; no real zeros. Confirm with the equation.

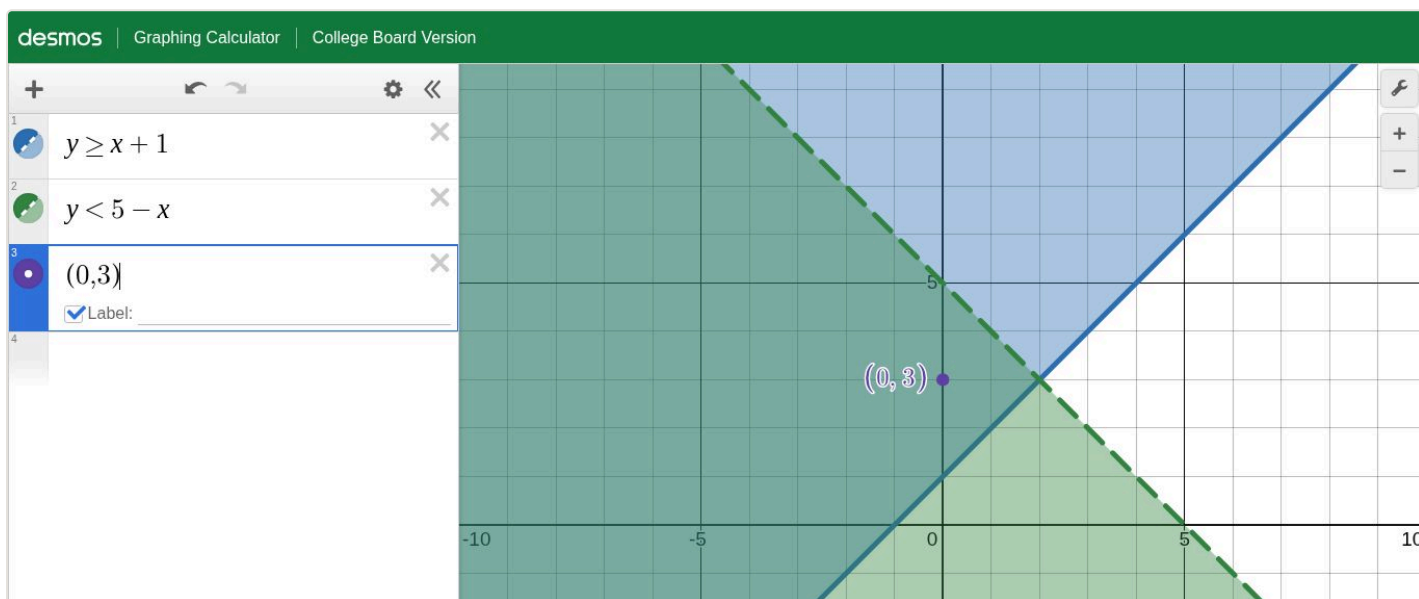
CALCULATOR LAB · 2 OF 4

## Desmos · regions & restrictions

Let the graph show the valid region; let the original conditions decide inclusion.

### 03 · Graph inequalities and find the overlap

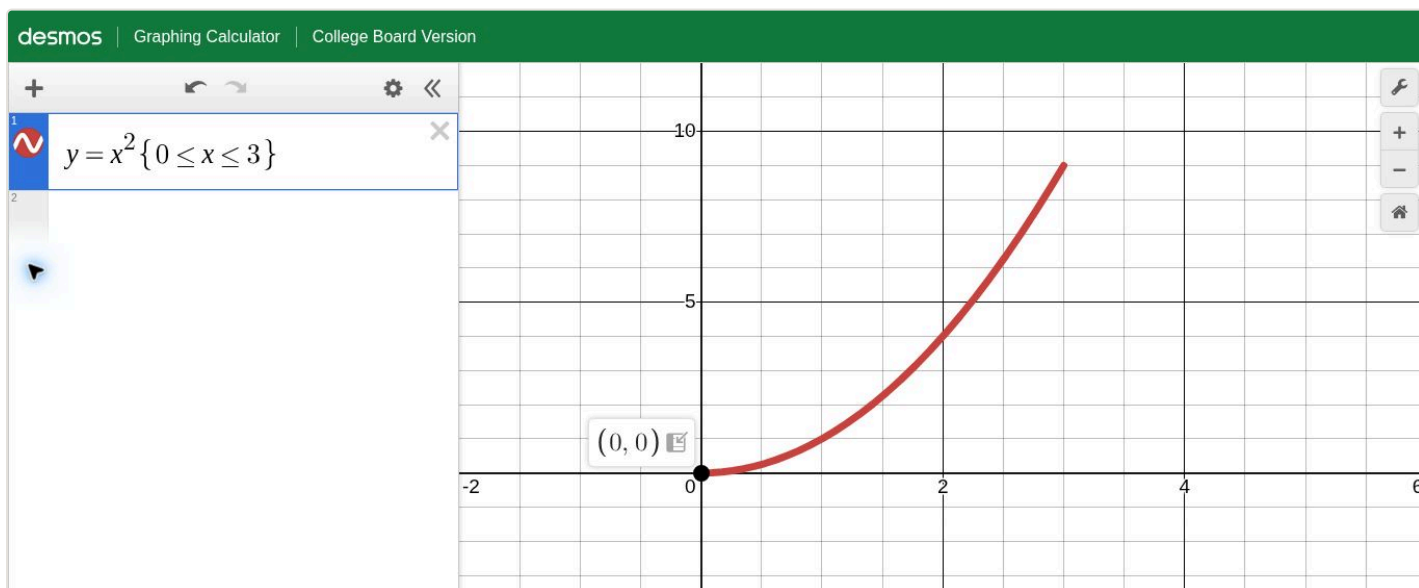
1. Enter  $y \geq x + 1$  and  $y < 5 - x$  on separate lines.
2. The overlap satisfies both inequalities. **Solid** boundaries include equality; **dashed** boundaries exclude it.
3. Check a candidate in both originals. The labeled point  $(0, 3)$  works:  $3 \geq 1$  and  $3 < 5$ .



For a one-variable inequality, you can enter it directly, such as  $x < 3$ . For multiple-choice points, substitution may be faster.

### 04 · Restrict the graph to allowed inputs

1. Put a restriction in braces after the expression:  $y = x^2 \{0 \leq x \leq 3\}$ .
2. Only the part with  $0 \leq x \leq 3$  is displayed. Here the endpoints are  $(0, 0)$  and  $(3, 9)$ .
3. Read the range on the restricted graph:  $0 \leq y \leq 9$ . Match strict or inclusive endpoints to the question.



**Window check:** seeing no intersection in one window does not prove there is none. Zoom out, consider the domain, and verify algebraically when needed.

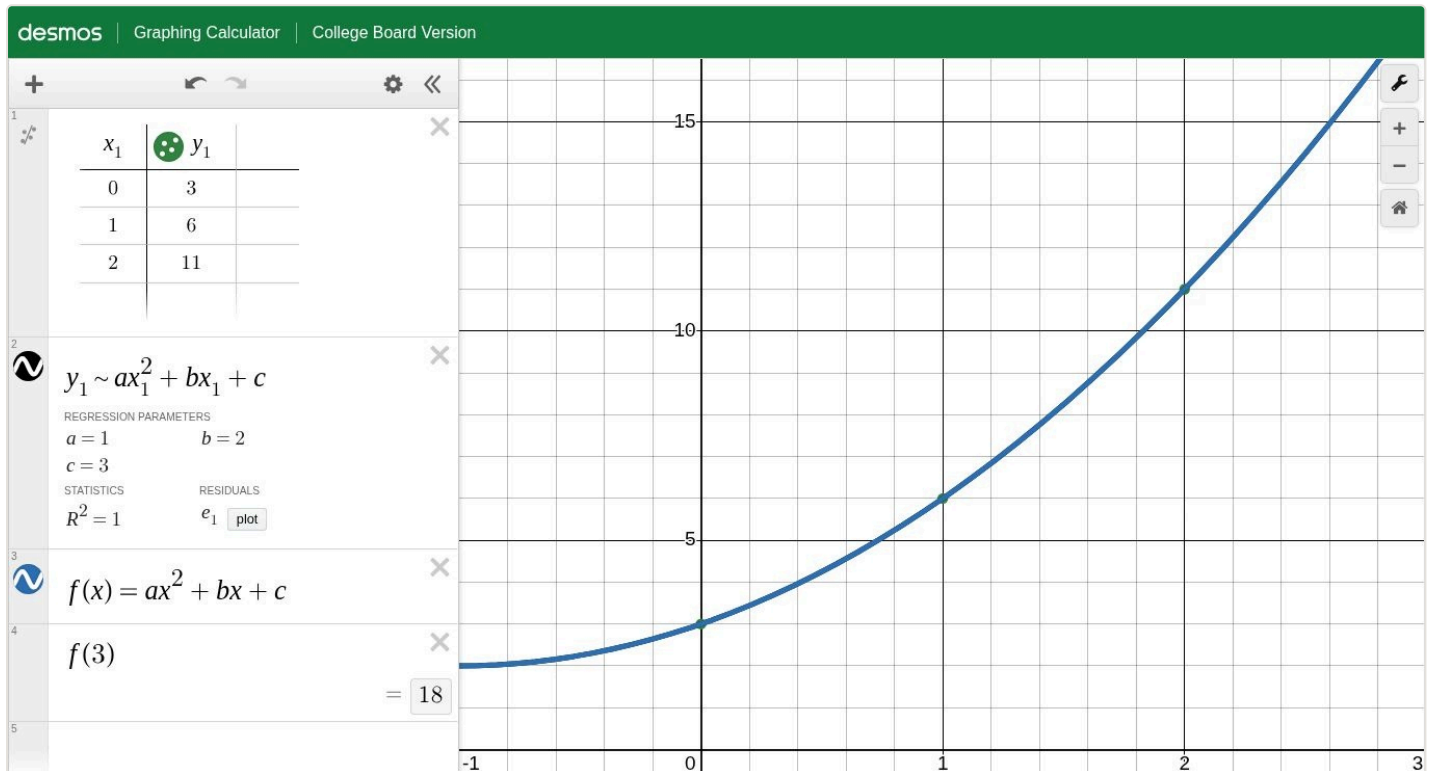
## CALCULATOR LAB · 3 OF 4

## Desmos · tables, models &amp; lists

Enter the data accurately. Use a model that matches the question.

## 05 · Use a table and fit a known model

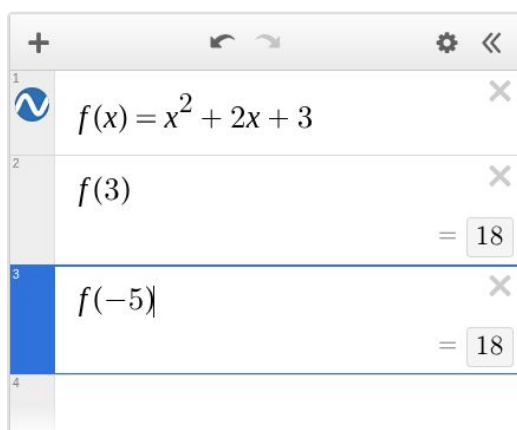
1. Choose + → **Table**. Enter  $x_1$ : **0, 1, 2** and  $y_1$ : **3, 6, 11**.
2. For a quadratic model, enter  $y_1 \sim a x_1^2 + b x_1 + c$ . Use the **tilde (~)**, not an equals sign.
3. Read  **$a = 1$ ,  $b = 2$ ,  $c = 3$** . Define  $f(x) = a x^2 + b x + c$ ; then  $f(3)$  gives **18**.



**Choose the family from the problem:** linear  $y_1 \sim m x_1 + b$ ; exponential  $y_1 \sim a b^{(x_1)}$ . A fit to a few points does not prove two expressions are identical.

## 06 · Evaluate a function

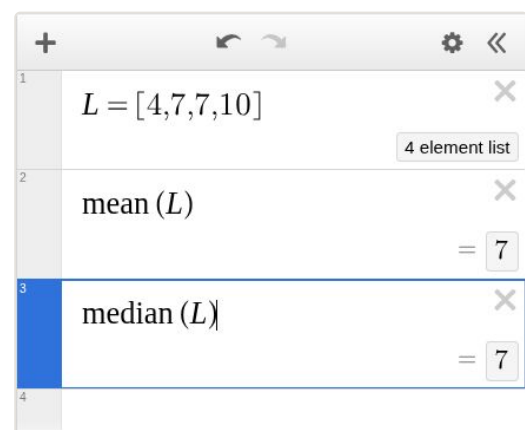
Define  $f(x) = x^2 + 2x + 3$ . On a new line, type  $f(3)$  or  $f(-5)$ .



**Read:** both inputs give 18. To solve  $f(x) = 18$ , instead graph  $y = f(x)$  and  $y = 18$ .

## 07 · Calculate with a list

Enter  $L = [4, 7, 7, 10]$ , then type  $\text{mean}(L)$  or  $\text{median}(L)$ .



**Read:** both are 7. Optional:  $\text{stdev}(L)$  is sample SD;  $\text{stdevp}(L)$  is population SD.

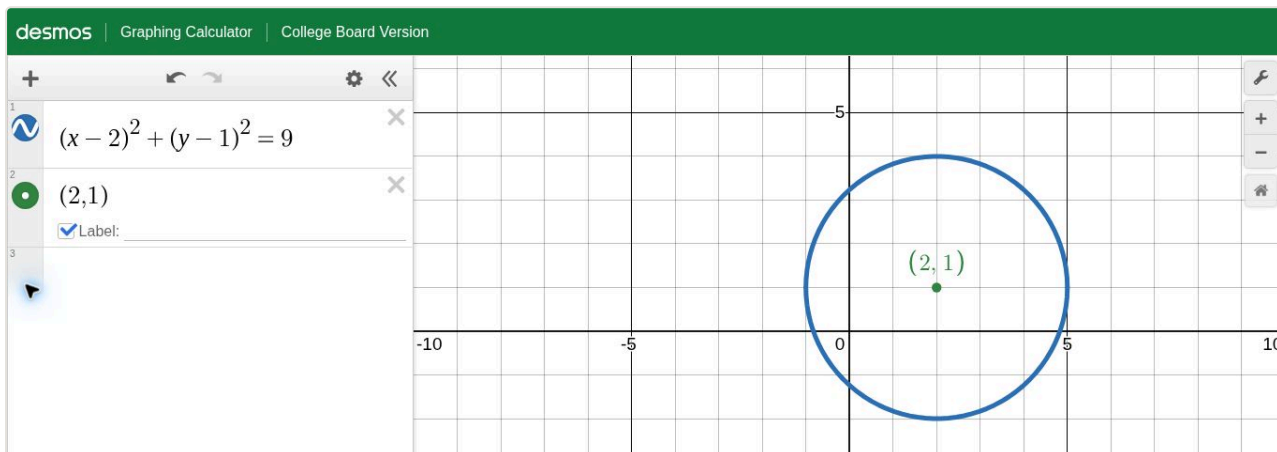
CALCULATOR LAB · 4 OF 4

## Desmos · circles, sliders & settings

Use the calculator to explore and calculate; use the mathematics to confirm.

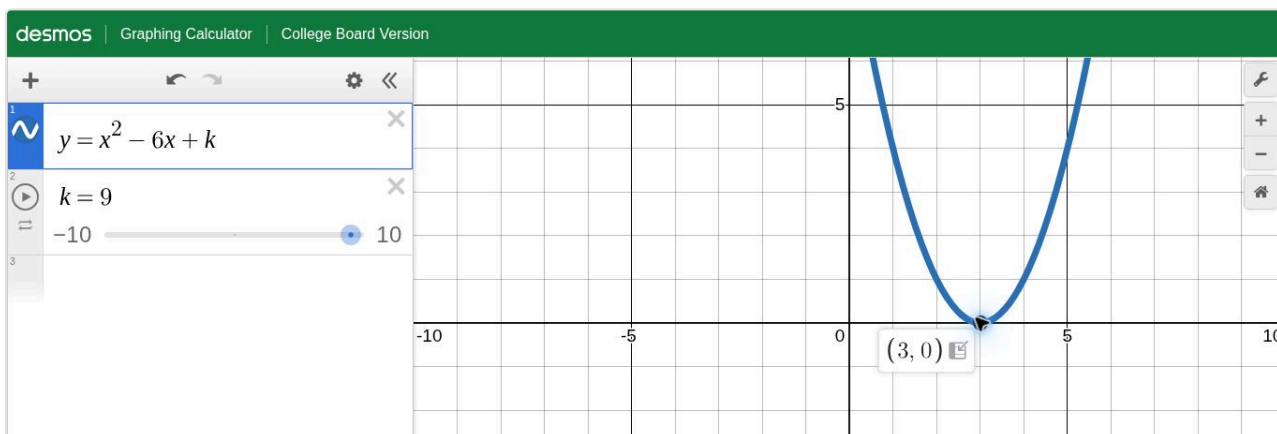
### 08 · Graph a circle and its center

Enter  $(x-2)^2 + (y-1)^2 = 9$ , then the point  $(2, 1)$ . The center is **(2, 1)** and the radius is **3**, not 9.



### 09 · Explore a parameter with a slider

Enter  $y = x^2 - 6x + k$ , then add a slider for  $k$ . Move it: at  $k = 9$ , the graph touches the x-axis at **(3, 0)**.



**Confirm the exact value:**  $D = 36 - 4k = 0 \rightarrow k = 9$ . A slider is useful for exploring a pattern; screen resolution cannot establish exact tangency.

### 10 · Set angle mode before using trig

Open the **wrench (Graph Settings)**, then choose **Degrees** or **Radians**. In Degrees mode,  $\sin(30)$  gives **0.5**.

Expression panel

1  $\sin(30)$  = 0.5

Graph Settings → angle mode

Radians Degrees

Two crops from the same calculator view; no mathematical results have been altered.

**Before accepting a calculator answer:** check parentheses, restrictions, window, angle mode, and the coordinate or quantity requested.

Reference: College Board math scope · College Board Desmos · regressions · statistics. Screenshots checked 7 September 2026. Independent study guide; not an official College Board publication.